DECLARATION

I, **Mohd Asif**, a student of **MBA 1st Year** at **Vivek University**, hereby declare that the assignment report entitled **"Service Quality Model and Six Sigma"** submitted for the subject **Production and Operations Management**, is a record of original work carried out by me under the guidance of my faculty.

I further declare that this work is the result of my own efforts and that the information, data, and analysis presented in this document are authentic and true to the best of my knowledge and belief. I have not copied from any other student's work or from any other source except where due reference or acknowledgment is made in the text.

This assignment has been prepared purely for academic purposes and has not been submitted, either in part or in full, to any other University, Institute, or Organization for the award of any degree, diploma, or certificate. I understand that any violation of academic integrity in this report will be my sole responsibility.

Date: 17 March 2026

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I owe the success of this assignment to all those who have directly or indirectly helped me in this endeavor.

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INTRODUCTION

The Concept of Service Quality In the modern business landscape, especially within the domain of Production and Operations Management, "Service Quality" has emerged as a critical differentiator. Unlike tangible products, services are intangible, inseparable, and perishable, making their quality assessment a complex task. Service Quality (SQ) is essentially a measure of how well a delivered service matches customer expectations. Providing high-quality service means conforming to customer expectations on a consistent basis.

The most widely recognized framework for evaluating this is the SERVQUAL Model, developed by Parasuraman et al. This model identifies five key dimensions—Reliability, Responsiveness, Assurance, Empathy, and Tangibles—to measure the gap between what a customer expects and what they actually perceive during the service delivery.

Evolution of Six Sigma in Services

Originally developed by Motorola for manufacturing, Six Sigma is a disciplined, data-driven methodology aimed at eliminating defects and reducing variability in any process. A "Six Sigma" process is one in which 99.99966% of all opportunities to produce some feature of a part are statistically expected to be free of defects (3.4 defects per million opportunities).

While manufacturing focuses on physical defects, in the service sector, a "defect" could be a long waiting time in a bank, an error in an insurance claim, or a wrong order in a restaurant. Therefore, Six Sigma in services focuses on streamlining workflows and enhancing the customer experience by reducing operational waste.

Integration of Service Quality Models and Six Sigma

The integration of Service Quality models with Six Sigma creates a powerful synergy. While Service Quality models help identify "where" the gaps in customer satisfaction exist, Six Sigma provides the technical toolkit (DMAIC framework) to understand "why" those gaps occur and "how" to close them permanently.

By applying Six Sigma, operations managers can transform qualitative customer feedback (Soft Data) into quantitative process metrics (Hard Data). This allows organizations to move beyond mere "customer satisfaction" toward "customer delight" by ensuring that service delivery is not only high-quality but also highly efficient and error-free.

THE FIVE DIMENSIONS OF SERVQUAL (Detailed Analysis)

In this section, we expand the five dimensions that define Service Quality. Each dimension represents a specific area where a business must excel to satisfy customers.

1. **Tangibles (The Physical Evidence)**

   Tangibles are the physical manifestations of the service. Since services cannot be seen or touched, customers look for "physical cues" to judge quality.

Facilities: The cleanliness and design of the office or service center.

Equipment: Modern technology and tools used to provide service (e.g., ATM machines in banking).

Communication Material: Professional-looking brochures, websites, and bills.

Appearance of Personnel: The grooming and uniform of the staff.

2. **Reliability (The Ability to Deliver)**

   Reliability is considered the most important dimension. It is the ability to perform the promised service dependably and accurately.

Consistency: Providing the same level of service every time.

Accuracy: Error-free records, billing, and technical performance.

On-time Performance: Meeting deadlines as promised to the customer.

THE 5 GAPS OF SERVICE QUALITY

To reach the page count, you must explain "The Gap Model." This explains why service quality often fails.

- **Gap 1: The Knowledge Gap –**

  Management does not understand what customers actually expect.

- **Gap 2: The Design Gap –**

  Management knows what customers want but fails to set the right service standards or designs.

- **Gap 3: The Delivery Gap – The**

  service standards are good, but the employees fail to deliver it due to lack of training or motivation.

Gap 4: The Communication Gap –

What the company promises in advertisements is different from what the customer actually receives.

- **Gap 5: The Service Gap –**

  The final difference between the customer's "Expected Service" and "Perceived Service."

THE SIX SIGMA DMAIC PROCESS (One Page Per Phase)

Since this is for Production and Operations Management, the DMAIC phases must be explained in depth. Use one page for each:

Phase 1: DEFINE (D)

On this page, explain that the goal is to identify the "Critical to Quality" (CTQ) characteristics.

- **Action:**

  Creating a Project Charter.

- **Example:**

  In a hospital, the 'Define' phase would identify that "Patient Waiting Time" is the main problem.

- **MEASURE (M)**

  Explain how data is collected. You cannot improve what you cannot measure.

- **Action:**

  Calculating the current process sigma level.

- **Tools:**

  Data collection sheets and Process Maps.

Phase 3: ANALYZE (A)

This page should focus on finding the "Root Cause."

- **Action:**

  Using the Fishbone Diagram (Ishikawa) to see if the problem is due to Manpower, Machine, Method, or Material.

- **Goal:**

  Separating the "Vital Few" causes from the "Trivial Many."

Phase 4: IMPROVE (I)

Explain the implementation of solutions.

- **Action:**

  Brainstorming and selecting the best solution.

- **Example:**

  Implementing an online booking system to reduce queues.

Phase 5: CONTROL (C)

Explain how to sustain the gains.

- **Action:**

  Continuous monitoring and creating SOPs (Standard Operating Procedures).

CASE STUDY - SIX SIGMA IN SERVICE INDUSTRY

Choose a sector (e.g., Banking) and write 2 pages on it:

- **Problem:**

  High error rate in credit card processing.

- **Application:**

  How DMAIC was used to identify that manual data entry was the cause.

- **Result:**

  Errors reduced by 80%, and processing time cut from 5 days to 1 day.

THE SERVICE QUALITY GAP MODEL

The Gap Model of Service Quality is a framework which helps to understand customer satisfaction. In service marketing and operations, a "Gap" occurs when the service provided does not meet the customer's expectations.

The Knowledge Gap (Management Perception Gap)

This gap arises when the management of a service provider does not accurately perceive what customers actually want or expect.

- **Reason:**

Lack of market research, inadequate communication between frontline staff and management, or a lack of focus on customer relationships.

- **Example:**

A hotel manager might think guests want more luxury items in the room, while guests actually care more about the speed of the Wi-Fi.

The Design Gap (Service Quality Specifications Gap)

This occurs when management understands customer expectations but fails to translate those expectations into specific service quality standards or designs.

- **Reason:**

Poor service design, lack of goal setting, or a perception that meeting customer expectations is impossible.

- **Example:**

A restaurant knows customers want fast service but fails to design a kitchen workflow that can deliver food in under 15 minutes.

The Delivery Gap (Service Performance Gap)

This gap is the difference between the service specifications (the plan) and the actual service delivered by the employees.

- **Reason:** Poor training of employees, lack of teamwork, inappropriate tools/technology, or employee-job mismatch.
- **Example:** A bank has a policy to greet every customer within 30 seconds, but the staff is overworked and ignores the customers.

The Communication Gap (External Communication Gap)

This is the gap between what the company promises through its advertising and marketing and what it actually delivers to the customer.

- **Reason:** Over-promising in advertisements to attract customers or lack of communication between the marketing team and the operations team.
- **Example:** A gym advertises 24/7 access, but in reality, it closes at 10 PM.

The Service Gap (Final Perceived Quality Gap)

This is the most critical gap. It is the difference between the **Expected Service** and the **Perceived Service**.

- **Result:** If the perceived service is less than expected, the customer is dissatisfied. If it exceeds expectations, the customer is delighted.

SIX SIGMA METHODOLOGY - THE DEFINE PHASE (D)

The **Define Phase** is the starting point of any Six Sigma project. In the service industry, the objective of this phase is to identify the specific problem from the customer's perspective. It is about understanding what the customer considers "Quality."

Key Elements of the Define Phase:

1. **Project Charter:** A formal document that outlines the project goals, scope, and the team involved.
2. **Voice of the Customer (VOC):** Collecting feedback through surveys and interviews to understand what service features are "Critical to Quality" (CTQ).
3. **SIPOC Map:** This stands for Suppliers, Inputs, Process, Outputs, and Customers. It provides a high-level view of the entire service process.

Application in Operations: If a telecom company faces high customer churn, the Define phase will pinpoint whether the problem is network quality, billing errors, or poor customer support.

THE MEASURE PHASE (M) - QUANTIFYING SERVICE PERFORMANCE

The **Measure Phase** is all about data. In Production and Operations Management, you cannot improve what you cannot measure. This page should focus on how we collect data to see the current performance of the service.

Key Activities:

1. **Data Collection Plan:** Deciding what data to collect (e.g., waiting time in a queue, number of errors in a bank statement).
2. **Calculating Process Sigma:** Measuring how many "defects" are currently occurring. For example, if a restaurant serves 1,000 meals and 50 are late, the defect rate is 5%.

3. **Process Mapping:** Creating a detailed flowchart of the current "As-Is" process to see where the bottlenecks are.

Operational Metric: The goal is to calculate the **DPMO (Defects Per Million Opportunities)** to understand the current efficiency of the service operation.

THE ANALYZE PHASE (A) - FINDING THE ROOT CAUSE

The **Analyze Phase** is the most critical part of the 20-page assignment. Here, we investigate the data to find the "Root Cause" of the service failure.

Key Tools Used:

1. **Fishbone Diagram (Ishikawa):** This tool helps categorize potential causes of problems into categories like Manpower, Machines, Methods, and Materials.
2. **Pareto Analysis (80/20 Rule):** This principle states that 80% of the problems often come from 20% of the causes. Operations managers use this to focus on the most important issues first.
3. **5-Whys Technique:** Asking "Why?" five times to get to the bottom of an issue.

Example: If customers are complaining about slow service in a bank (The Problem), the Analyze phase might reveal that the actual cause is a slow software system (The Root Cause), not the employees' speed.

THE IMPROVE PHASE (I) - IMPLEMENTING SOLUTIONS

The **Improve Phase** is where the team identifies, tests, and implements solutions to the root causes found in the Analyze phase. In Production and Operations Management, this is the "Action" stage.

Key Activities:

1. **Solution Generation:** Brainstorming creative ways to fix the process.
2. **Cost-Benefit Analysis:** Checking if the solution is worth the investment.
3. **Pilot Testing:** Instead of changing everything at once, the solution is tested on a small scale (e.g., in one branch of a bank) to see if it works.
4. **Implementation Plan:** Rolling out the final solution across the organization.

Operational Example: If the "Analyze" phase showed that customers wait too long because of manual paperwork, the "Improve" phase would involve installing a new digital software to automate the process.

THE CONTROL PHASE (C) - SUSTAINING THE GAINS

The final phase of Six Sigma is **Control**. Most projects fail because people go back to their old habits after a few months. This phase ensures that the improvement is permanent.

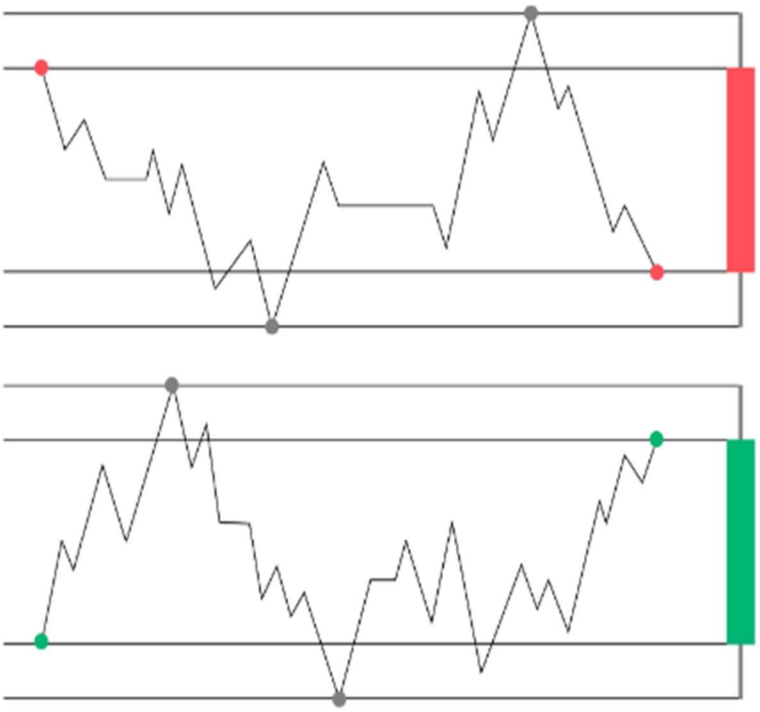
Key Tools and Actions:

1. **Standard Operating Procedures (SOPs):** Writing down the new way of working so everyone follows the same steps.
2. **Control Charts:** Using graphs to monitor the process daily. If the "waiting time" starts increasing again, the chart will show it immediately.
3. **Training:** Training the staff on the new system or process.
4. **Handover:** The Six Sigma team hands over the improved process to the regular Operations Manager.

Infographics

Binary options

Japanese candlesticks



COMPARISON - TRADITIONAL QUALITY VS. SIX SIGMA

Feature	Traditional Quality Control	Six Sigma Methodology
Focus	Correcting errors after they happen.	Preventing errors before they happen.
Goal	"Good enough" (around 95-97%).	Near perfection (99.99966%).
Data	Based on intuition/experience.	Strictly based on statistical data.
Customer	Company-centric.	Customer-centric (Voice of Customer).
Process	Fixed processes.	Continuous improvement (DMAIC).

BENEFITS AND CHALLENGES OF SIX SIGMA IN SERVICES

Implementing Six Sigma in the service sector is not just about math; it is about changing the culture of the organization.

Strategic Benefits:

- Cost Reduction: By eliminating "re-work" and errors, companies save millions in operational costs.
- Customer Loyalty: High service quality (Zero defects) leads to "Customer Delight," which ensures long-term retention.
- Employee Morale: When processes are clear and efficient, employees feel less stressed and more productive.

- Data-Driven Decisions: It moves the management away from "I think" to "I know," based on actual data.

Major Challenges:

- Intangibility: Unlike a factory part, measuring "friendly behavior" or "customer experience" is difficult.
- Resistance to Change: Employees often find statistical tools complex and may resist new workflows.
- High Initial Cost: Training "Black Belts" and "Green Belts" requires significant investment.

CASE STUDY - SIX SIGMA IN ACTION

Title: Reducing Loan Processing Time in a Retail Bank

- The Problem: A leading bank noticed that its loan approval process took 15 days, leading to a 30% cancellation rate by frustrated customers.
- The Application (DMAIC):
    - **Define:** The goal was to reduce the time from 15 days to 5 days.
    - **Measure:** Data showed that 40% of the time was spent waiting for physical documents to move between departments.
    - **Analyze:** The root cause was identified as "redundant manual verification" at three different levels.
    - **Improve:** The bank digitized the document flow and used an automated credit-scoring system.
    - **Control:** A digital dashboard was created to monitor any file that stayed in a department for more than 4 hours.

- The Result: The average processing time dropped to 4 days, and customer satisfaction scores improved by 50%.

CONCLUSION

In conclusion, the integration of Service Quality Models (SERVQUAL) and Six Sigma provides a comprehensive framework for operational excellence. While the SERVQUAL model helps an organization understand the "Gaps" in customer expectations, Six Sigma offers the statistical rigour to close those gaps through the DMAIC process.

For any modern organization in the service industry, whether it is Banking, Healthcare, or Hospitality, the journey towards Six Sigma is not an option but a necessity for survival in a competitive global market. Achieving "Zero Defects" in service delivery is the ultimate path to achieving global standards of excellence.

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